

Using Digitized Herbarium Data in Research: Applications for Exploration, Taxonomy, and Ecology

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University of Florida

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iDigBio
Integrated Digitized Biocollections

NSF



Topics

- Intro to data
- Downloading data
- Cleaning data
- Georeferencing
  - GEOLocate
- Ecological Niche Modeling
  - Maxent
- ENM Analysis & Interpretation
- Uses of collections data
- Discussion/Q&A
- R-based methods

Workshop Leaders

- Pam Soltis
- Makenzie Mabry
- Shelly Gaynor
- Cameron McMullen
- Sarah Ellen Strickland
- Elizabeth White
- JT Miller
- Doug Soltis
- Contributions from many previous lab members/colleagues:
  - Blaine Marchant, Charlotte Germain-Aubrey, Andre Naranjo, Anthony Melton, Tal Kinser, Maria Kinser, Mike Belitz, Rhett Rautsaw, João Vidal, Jr.

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iDigBio
Integrated Digitized Biocollections



Logistics

- HiPerGator – UF's supercomputer
- GitHub repository
  - data
  - scripts
  - presentations
- Workshop evaluation survey
  - by email
- Pace/breaks


FLAS 22883
R. K. Godfrey Herbarium (FSU)
HERBARIUM
177581
FLORIDA STATE UNIVERSITY
The New York Botanical Garden
Woody Plants of U. S. A.
ACERACEAE Acer rubrum L. Female
Small tree, 8 m. tall, with two stems from base, 15 cm. in diameter.
16709-a April 22, 1979. Tree in full bloom, exclusively female (see no. 16708, a nearby male tree). Tips of calyx and corolla lobes light red as in no. 16708. Stigmas bright red.
16709-b June 4, 1979. Leaves dark green above, pale green below. Fruits pale tan.
Common in bottom of hollow, ungrazed woods of Quercus alba, Fraxinus americana, Populus grandidentata, Carya ovata.
WISCONSIN: Richland Co., 3 miles SE of Richland Center.
T 10 N; R 1 E; SE¼SW¼ sec. 25. alt. 250 m.
Coll.: M. Nee
HERBARIUM
UNIVERSITY OF FLORIDA
GAINESVILLE
135638

Herbaria Worldwide

NYBG STEERE HERBARIUM

Home Collections Discover Index Herbariorum Virtual Herbarium Loans Digitization



Index Herbariorum

3,400 herbaria

400,000,000 specimens!



Natural History Collections



1-2 billion specimens
in the US
3-4 billion specimens
worldwide

~1600 natural history collections in the US, thousands in the world



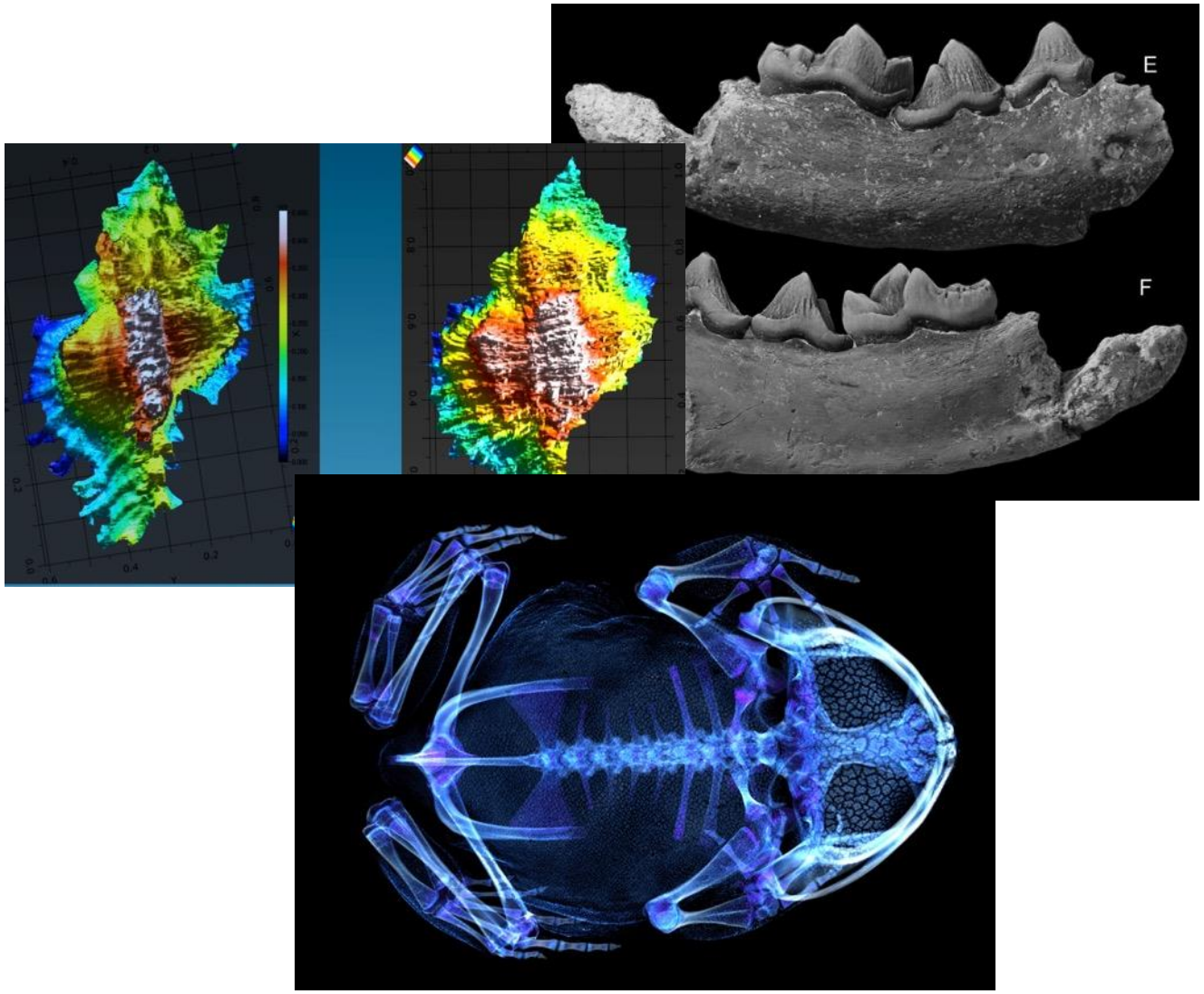
Smithsonian

Digitization: Data and Images

- Scientific name –
  including authority
- Date
- Collector
- Location – state, county,
  specific site,
  GPS coordinates
- Associated species
- Notes

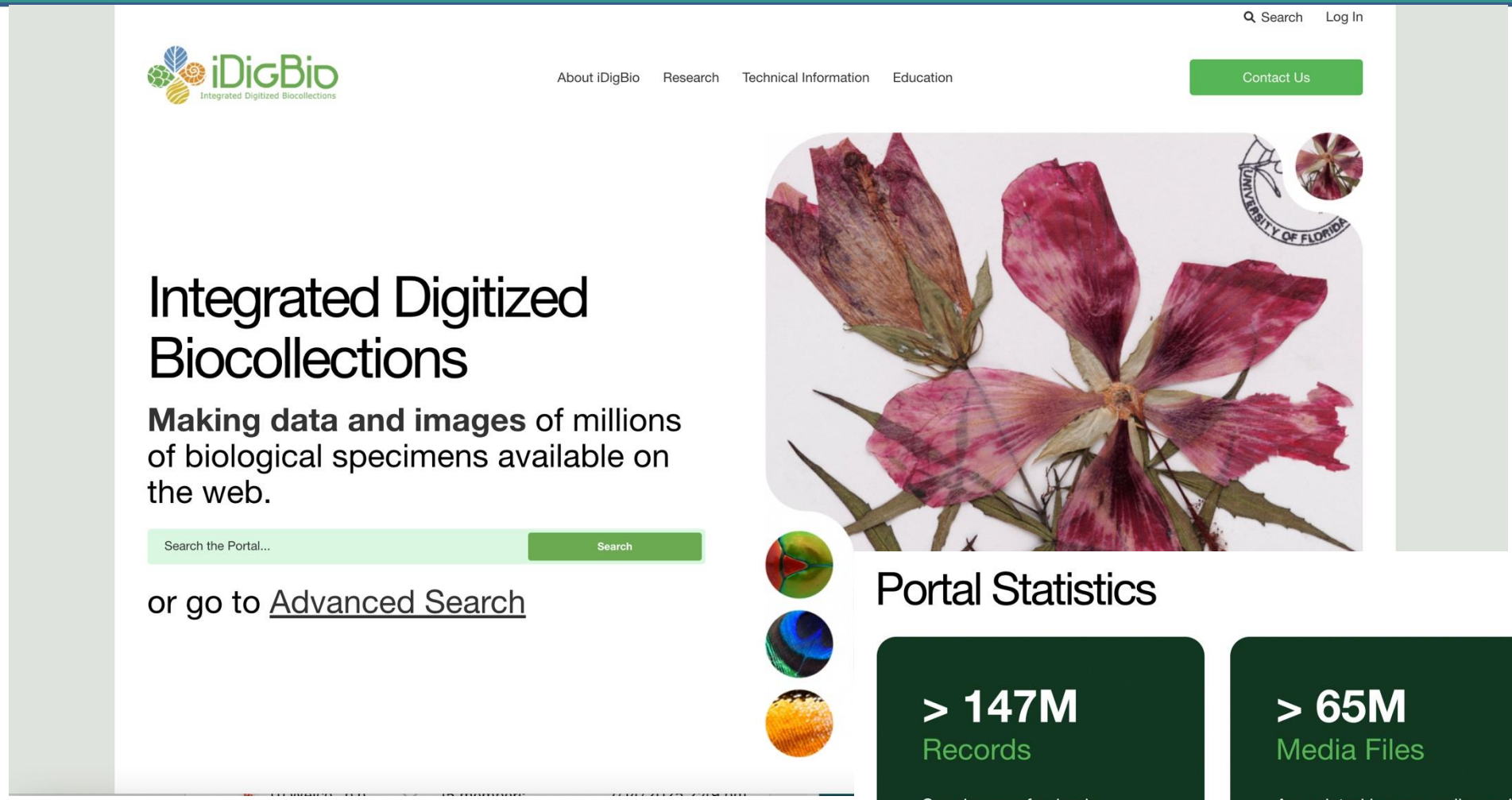


Digitization: Data and Images





iDigBio: A Source for Specimen Records



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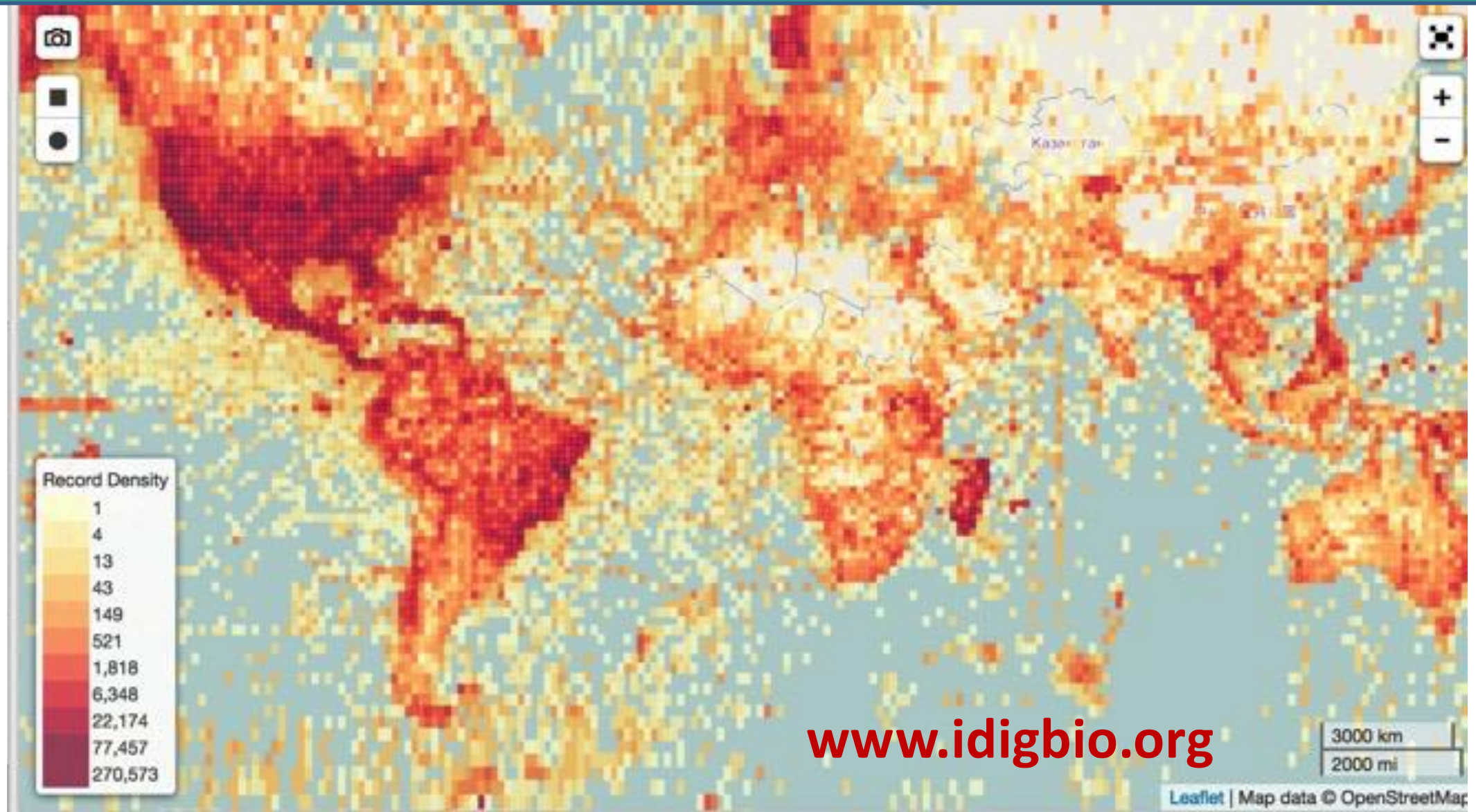
Portal Statistics

> 147M
Records
Specimens of animals, plants, fungi, other...

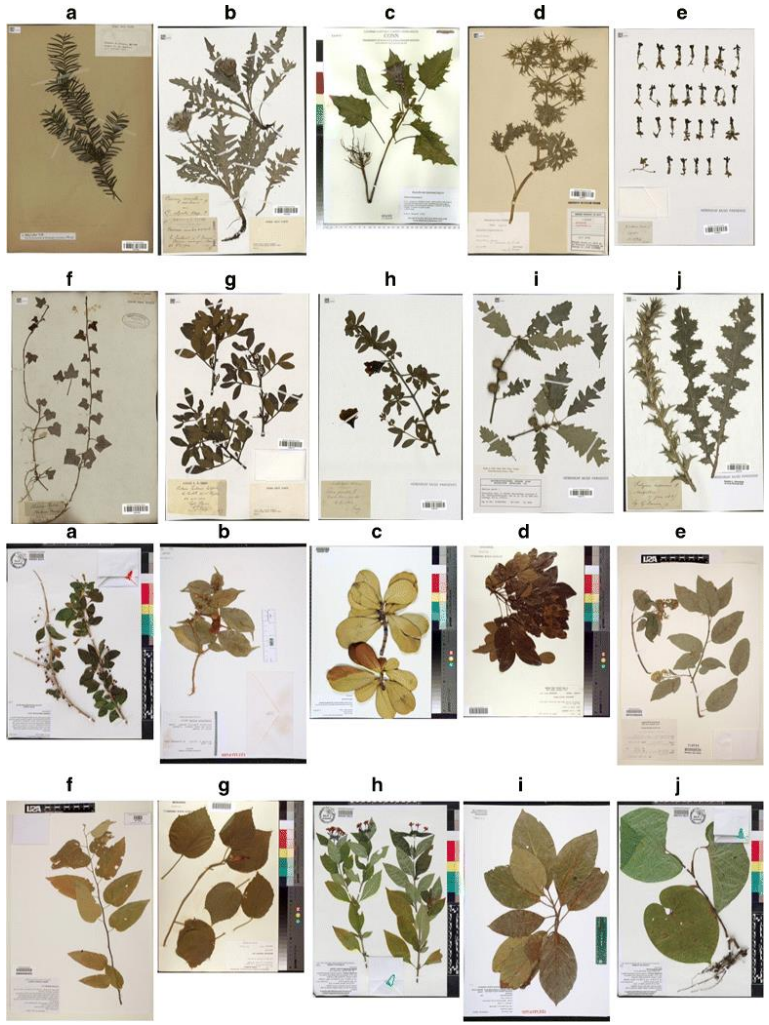
> 65M
Media Files
Associated image, audio, and video files

> 1.9K
Recordsets
Data from biodiversity collections in the US

Specimen Occurrences in iDigBio



Machine Learning & Biodiversity Research

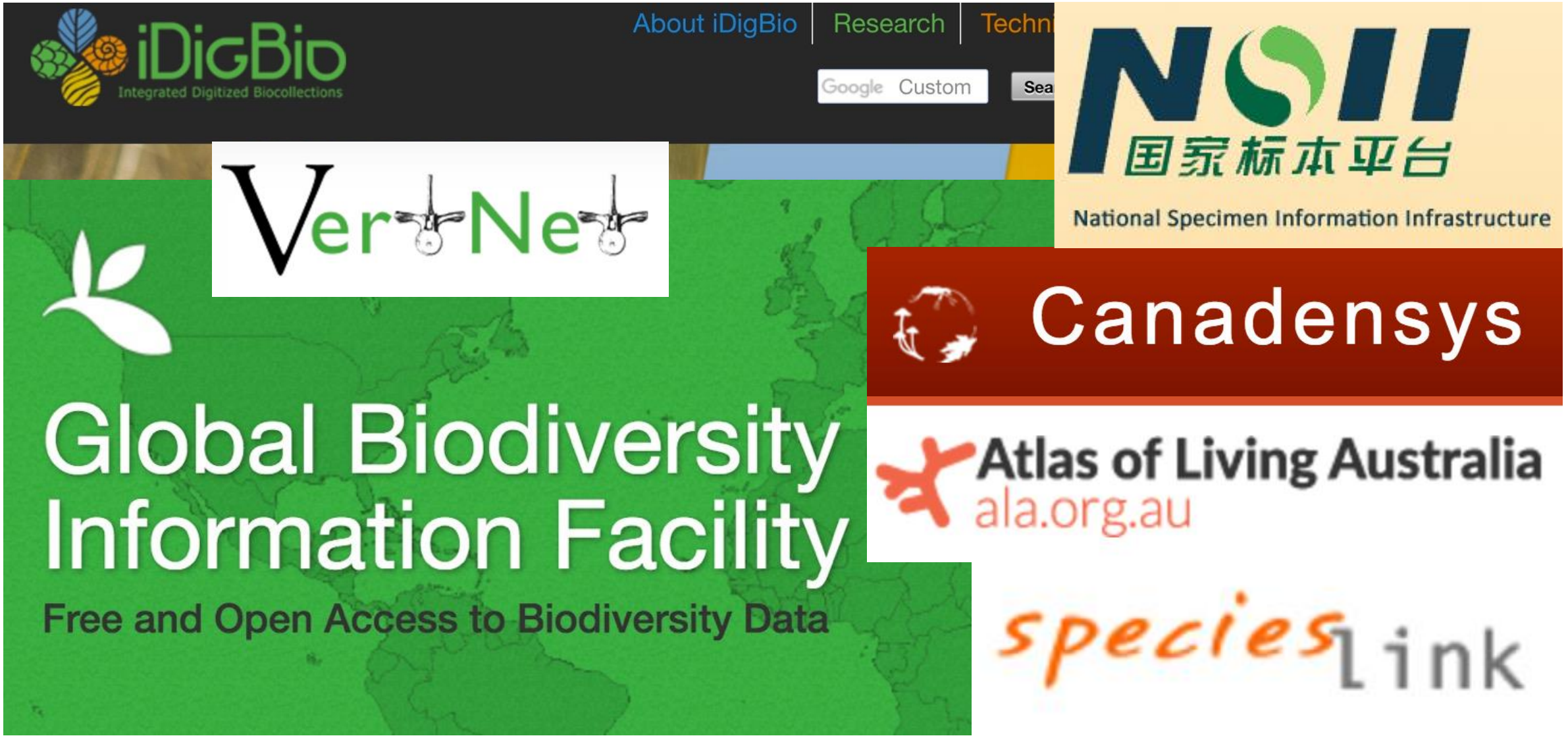


Carranza-Rojas et al. 2017



Pearson et al. 2020

Other Data Aggregators



‘Big Data’ Research in Biodiversity Science

- Monitoring shifts in biodiversity
- Tracking invasive species
- Ecological Niche Modeling, climate change
- Past movements and climate change
- Tracking phenological shifts
- Landscape genetics
- Integration of ENM with phylogeny
- Community phylogenetics/assembly
- Biogeography



Overview of Day's Activities

8:00	Welcome and Overview of the Workshop – Pam
	Darwin Core and Data Fields – Pam
	Data Downloading Overview – Pam
8:20	***Activity:*** Manual Data Downloads from iDigBio Portal – Pam
8:30	Intro to HiPerGator – Makenzie
8:35	Data Downloads – Shelly
8:45	***Activity:*** R-based Data Downloads – Shelly
9:00	Data Cleaning – Cameron
9:15	***Activity***: Data Cleaning
9:30	Georeferencing – Sarah Ellen
10:00	**Coffee Break**
10:30	Environmental Variables & Data Exploration – Sarah Ellen
10:45	***Activity:*** Environmental Variables & Data Exploration – Sarah Ellen

Overview of Day's Activities

11:15	**Lunch**
12:15	Defining Accessible Area + Variable Selection – Makenzie
12:25	***Activity***: Defining Accessible Area + Variable Selection – Makenzie
12:45	Ecological Niche Models (ENMs) – Elizabeth
1:00	***Activity***: ENMs – Elizabeth
1:30	Null Models – JT
1:40	***Activity***: Null Models – JT
2:00	Post ENM analyses + Future Predictions – Elizabeth
2:15	***Activity***: Post ENM analyses + Future Predictions – Elizabeth
2:45	Phylodiversity and Phyloendemism – JT
3:00	**Coffee Break**
3:30	***Activity***: Phylodiversity and Phyloendemism – JT
3:45	Closing with Examples – Doug
4:00	Final Q & A